

Host-Pathogen Interactome: Biocuration and Computational Prediction

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Introduction

Host-pathogen interactions play a vital role in initiating and maintaining infections. However, many diseases have little or no available HPis in existing molecular interaction databases, especially for animal pathogens. Therefore, resources that annotate, predict and display the HPI are critical for developing novel intervention strategies.

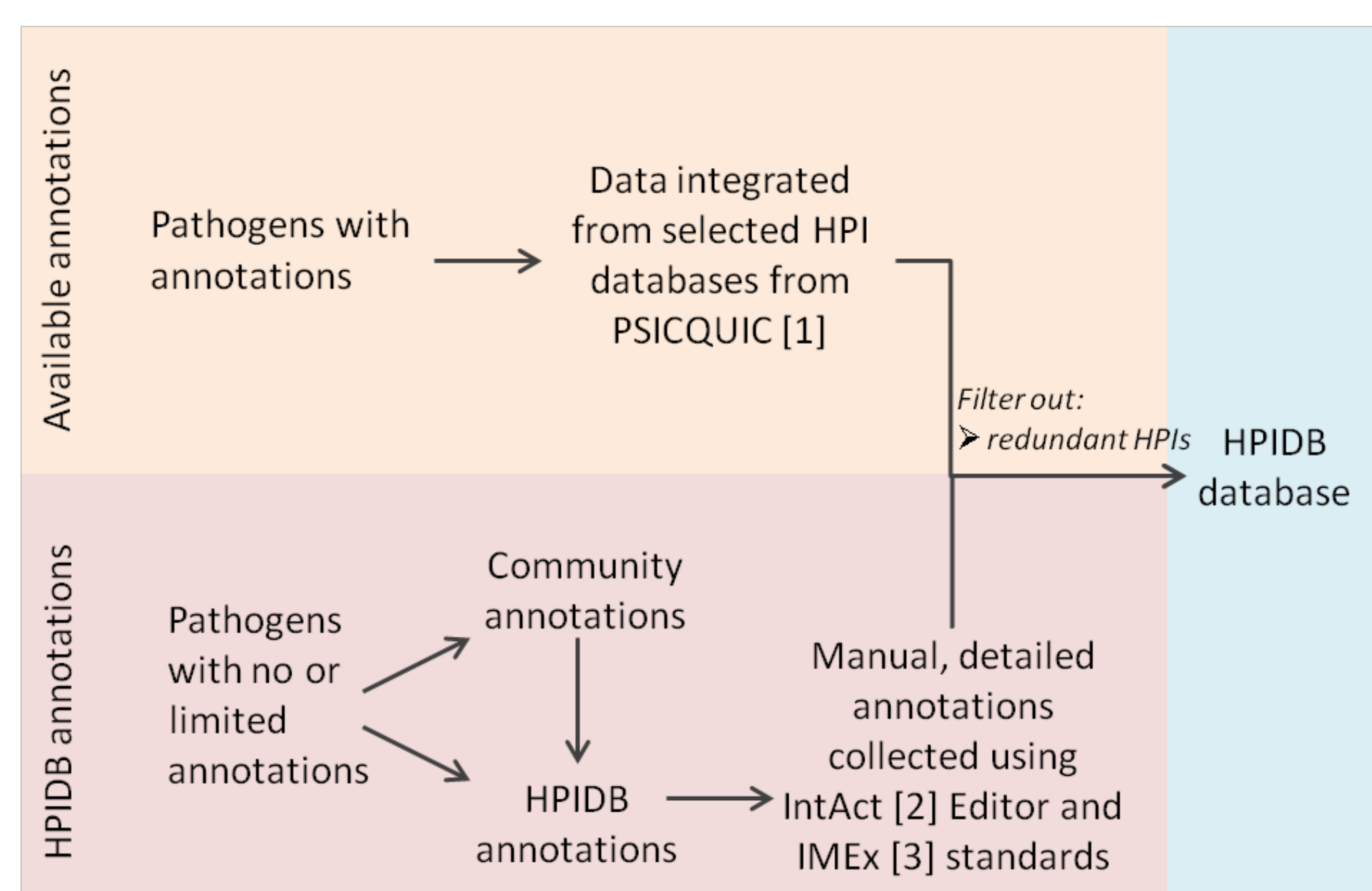
The *Host Pathogen Interaction Database (HPIDB)* provides:

- (1) access to high quality, comprehensive HPI data from a broad range of pathogens and their hosts,
- (2) ability to transfer HPI information across multiple host-pathogen systems and
- (3) network visualization and functional analysis of results.

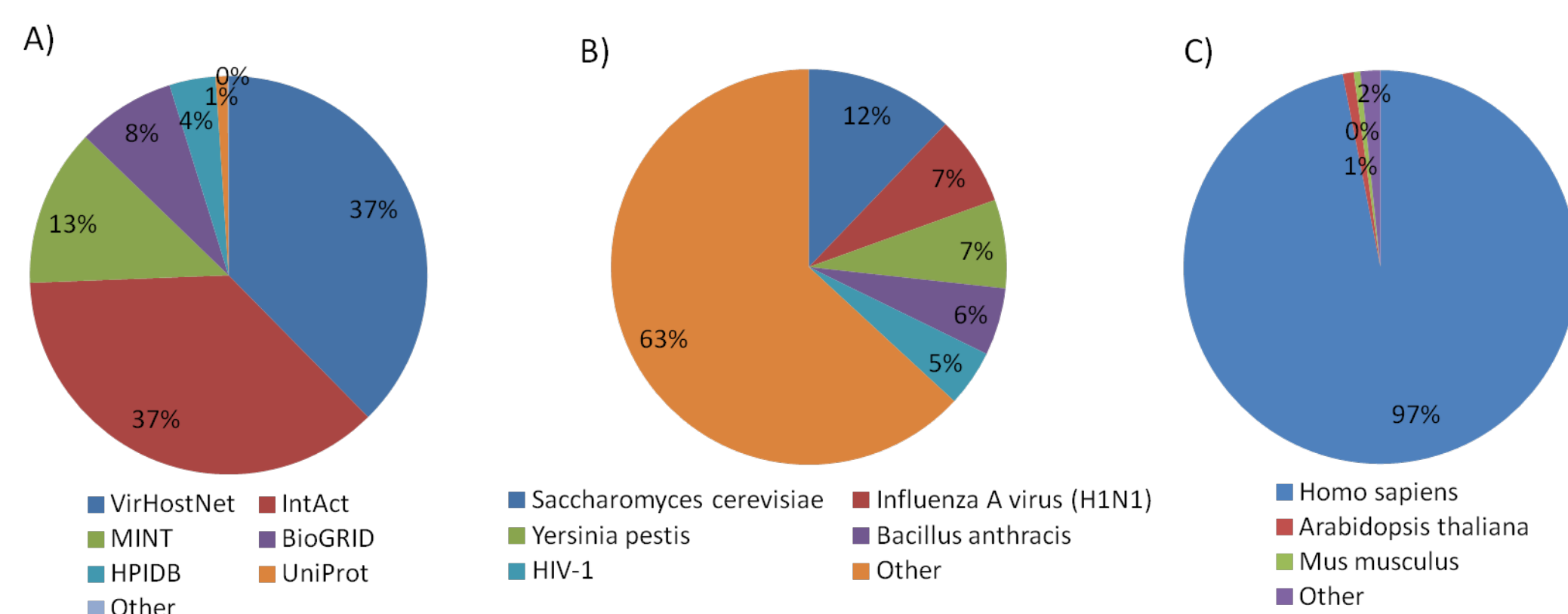
All of HPIDB data and tools are freely available and is updated on a regular basis.

URL: <http://agbase.msstate.edu/hpi/main.html>

HPI Annotations



HPIDB datasets. HPIDB provides targeted manually curated annotations that are complemented by integrating existing HPI information from external databases.



HPIDB statistics. Distribution of HPis in HPIDB based on (A) databases, (B) pathogen and (C) host species. As of March 2017, HPIDB contains 55,505 entries which include 55 host and 523 pathogen species.

To request HPI annotations & assistance:
agbase@igbb.msstate.edu

HPI Prediction

HPIDB “Search Homologous HPis” module. This module allows researchers to enter their sequences to find homologous pairs of interacting proteins from different organisms.

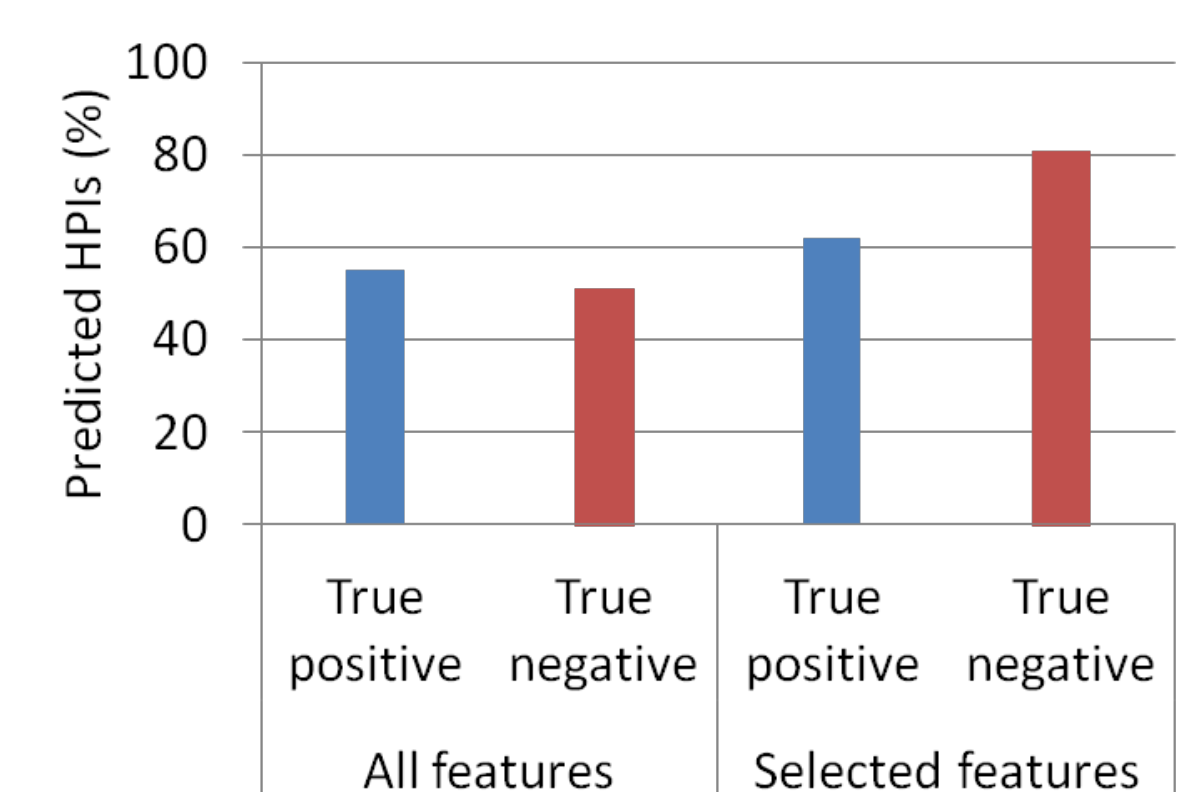
A)

Method	All features			
	Accuracy	Precision	Recall	F1 Score
Random Forest	83	60	59	59
Naïve Bayes	64	59	59	59
SVM	53	61	51	36

B)

Method	Selected features			
	Accuracy	Precision	Recall	F1 Score
Random Forest	82	54	52	50
Naïve Bayes	64	59	59	59
SVM	53	62	51	36

Performance evaluation of Machine Learning model. Comparison of the percentage of accuracy, precision, Recall and F1 scores for viral test data when using a training file with (A) All or (B) selected features and three different classifiers. Similar work is done for other pathogen categories (data not shown).



Overview of prediction model process. For improved computational HPI prediction, we are developing a Machine Learning workflow so that researchers can rapidly predict and model interactions for their own pathogens of interest.

Ability to recover known HPis. Percentage of experimentally verified Classical swine fever virus-host interactions that are predicted using Random Forest classifier trained on a mix of viruses with All or Selected features.

Future Directions

➤ We continue to develop HPI prediction workflows, evaluating different methods, protein features, and assessment. Our goal is to establish our prediction workflow as an online tool so that researchers can rapidly predict and model interactions for their own pathogens of interest in a genome wide scale.

➤ HPIDB manual annotation will focus on a broad range of pathogen systems, and will serve as a core “gold standard” set to predict similar HPI in related host-pathogen systems and to test the accuracy of these predictions.

➤ *We are interested in working with researchers who are interested in HPI analysis of their own data sets.*

References

- [1] PSICQUIC: <http://www.ebi.ac.uk/Tools/webservices/psicquic/view/main.xhtml>
- [2] Orchard S, et al. Nucleic acids research, 42, D358-363 (2014)
- [3] Orchard S, et al. Nature methods, 9(4):345-350 (2012)